

Decode Labs

Cyber Security Internship

PROJECT 3

Phishing Awareness Analysis

Submitted By
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Decode Labs Internship
Batch 2026

Introduction

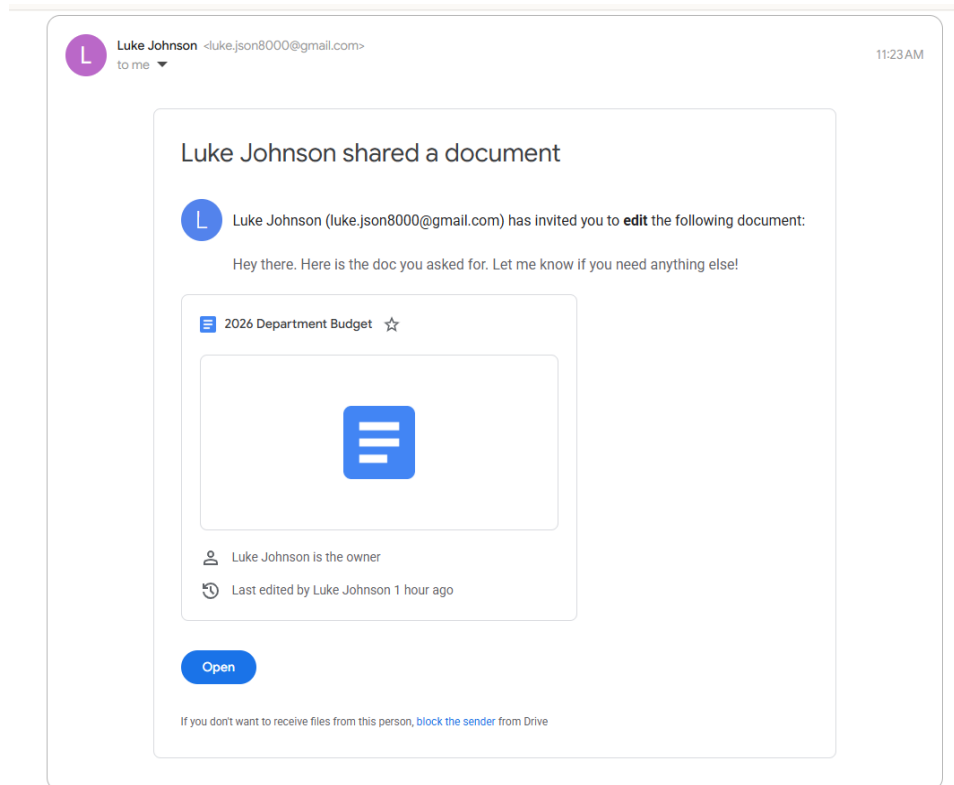
Phishing is one of the most common and dangerous cyberattacks used by cybercriminals to deceive users into revealing sensitive information such as usernames, passwords, financial details, and personal data. Attackers often disguise malicious emails as legitimate communications from trusted organizations by using convincing branding, urgent language, and deceptive links.

The objective of this project is to analyze suspicious email messages and identify indicators that suggest phishing attempts. Each email is examined from both a technical and behavioral perspective by evaluating the sender's address, domain, message content, social engineering techniques, and potential security risks.

Through this analysis, I developed practical skills in identifying phishing indicators, assessing potential threats, and recommending appropriate security measures. These skills are essential for cybersecurity professionals responsible for protecting individuals and organizations from email-based attacks.

Email 1 Analysis

Screenshot



Overall Verdict

This email displays several indicators commonly associated with phishing and credential-harvesting attacks. Although it appears to be a legitimate Google Drive sharing notification, the sender information and message context raise significant security concerns. The email should be treated as suspicious until its authenticity is verified through a trusted communication channel.

1. Sender Domain vs. Content Misalignment (Primary Red Flag)

Indicator: The sender uses a personal Gmail address (luke.json8000@gmail.com) to share a document titled “2026 Department Budget.”

Analysis: Department budgets and other confidential business documents are typically shared through an organization’s official email domain. Receiving such sensitive information from a personal Gmail account is unusual and may indicate an attempt to impersonate a trusted individual.

Risk: The attacker may be attempting to gain the recipient’s trust by using a familiar name while avoiding the security controls associated with a legitimate corporate email account.

2. Social Engineering Through Curiosity and Urgency

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Indicator: The shared document is titled “2026 Department Budget.”

Analysis: Cybercriminals frequently use subjects related to finances, salaries, budgets, invoices, or company announcements because they naturally attract attention.

Risk: Victims may click the shared document impulsively, increasing the likelihood of credential theft or malware infection.

3. Vague and Familiar Language

Indicator: The message states: “Here is the doc you asked for.”

Analysis: This wording assumes that the recipient previously requested the document. Such vague language is a common social engineering tactic designed to create familiarity and exploit uncertainty.

Risk: Recipients may question whether they forgot requesting the document and click the link simply to verify.

4. Use of a Trusted Service

Indicator: The email appears to originate from a Google Drive sharing notification.

Analysis: Attackers frequently abuse trusted cloud platforms such as Google Drive, Microsoft OneDrive, or Dropbox because emails from these services often appear legitimate and may bypass spam filters.

Risk: Although the notification may appear genuine, the shared file or destination could still be malicious.

5. Suspicious Call-to-Action

Indicator: The email encourages the recipient to click the “Open” button.

Analysis: Unexpected requests to open documents or click links should always be treated with caution.

Risk: The button could redirect users to a malicious website or a fake login page designed to steal usernames and passwords.

Red Flags Identified

- Personal Gmail address used to share confidential business information.
- Sensitive document title intended to attract attention.
- Unexpected document sharing request.
- Vague message that assumes previous communication.
- Reliance on a trusted cloud service to appear legitimate.
- Encourages immediate interaction through the “Open” button.
- Sender identity cannot be verified from the email alone.

Risk Assessment

Category	Assessment
Threat Type	Phishing / Credential Harvesting

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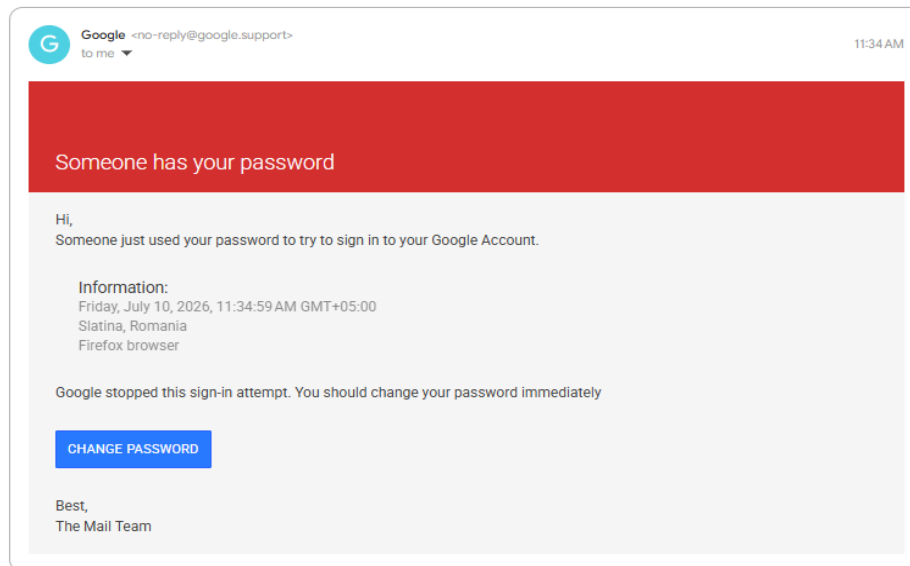
Category	Assessment
Risk Level	High
Potential Impact	Account compromise, malware infection, data theft
Confidence Level	Medium-High

Recommended Actions

- Do not click the “Open” button or any links contained in the email.
- Verify the sender through a trusted communication channel such as Microsoft Teams, Slack, or a known phone number.
- Report the email to your organization’s IT or Security team using the official phishing reporting process.
- Inspect the sender’s email address carefully before trusting the message.
- Enable Multi-Factor Authentication (MFA) to reduce the impact of stolen credentials.
- Delete the email if it is confirmed to be malicious.

Email 2 Analysis

Screenshot



Overall Verdict

This email contains several indicators commonly associated with phishing attacks. The sender's domain, branding inconsistencies, and psychologically persuasive language raise concerns about its authenticity. The email should be considered suspicious until verified through official Google security channels.

1. Header and Email Domain Discrepancies (Primary Red Flag)

Indicator: The email appears to be sent from Google <no-reply@google.support>.

Analysis: Official Google security notifications are typically sent from trusted Google-owned domains such as @google.com or @accounts.google.com. The use of the .support top-level domain instead of Google's standard domains is unusual and should be treated with caution.

Risk: A deceptive sender domain may be used to impersonate Google and increase the likelihood that recipients trust the email.

2. Branding and Formatting Inconsistencies

Indicator: The email contains branding elements that differ from typical Google security notifications.

Analysis: Legitimate security emails from Google generally follow a consistent design, language, and branding style. Features such as the sign-off "Best, The Mail Team" or outdated design elements may indicate that the email does not originate from Google's official communication system.

Risk: Branding inconsistencies can be signs of impersonation intended to convince recipients that the email is genuine.

3. Social Engineering Techniques

Indicator: The email states that someone attempted to access the account from an unfamiliar location and urges the recipient to change their password immediately.

Analysis: This message uses fear and urgency to encourage immediate action. Mentioning an unknown location and a potential account compromise is intended to trigger an emotional response before verification.

Risk: Recipients may respond impulsively instead of verifying the alert through official account settings.

4. Suspicious Call-to-Action

Indicator: The email includes a prominent “CHANGE PASSWORD” button.

Analysis: Unexpected password reset requests should always be treated cautiously. If the email is malicious, the button could redirect users to a fraudulent website designed to imitate Google’s login page.

Risk: Victims may unknowingly disclose their Google account credentials to attackers.

Red Flags Identified

- Suspicious sender domain (@google.support).
- Possible lookalike domain impersonating Google.
- Unusual branding and email signature.
- Fear-based messaging.
- Strong sense of urgency.
- Prominent password reset button.
- Requests immediate action without independent verification.

Risk Assessment

Category	Assessment
Threat Type	Phishing / Credential Harvesting
Risk Level	High
Potential Impact	Account compromise, identity theft, unauthorized account access
Confidence Level	High

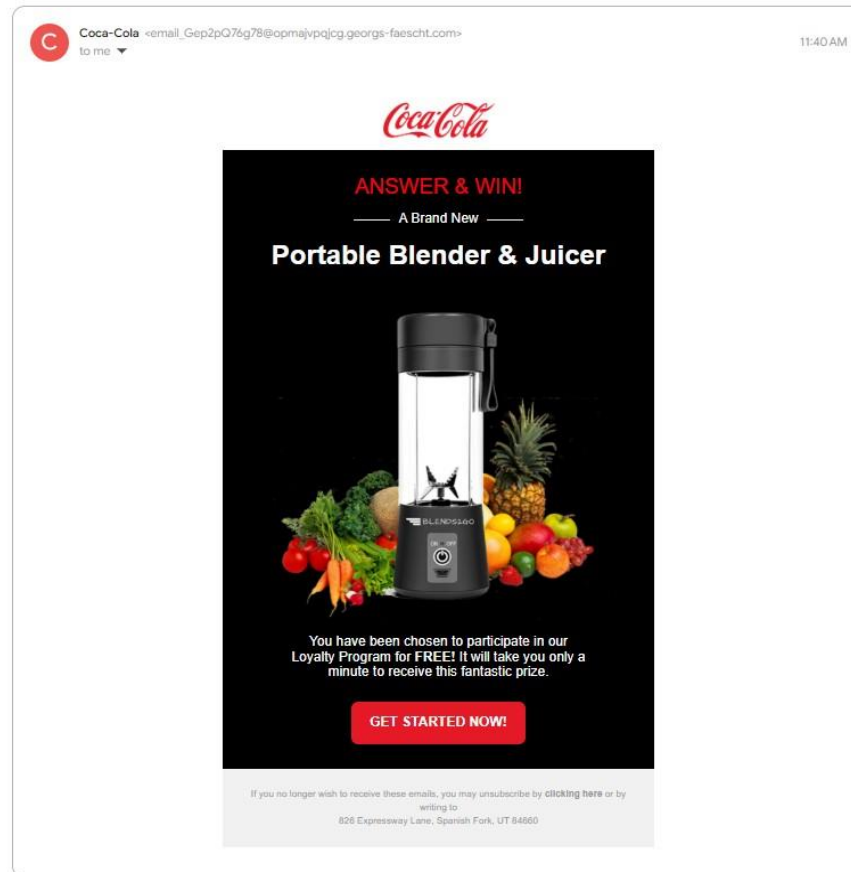
Recommended Actions

- Do not click the “CHANGE PASSWORD” button.
- Verify the alert by manually visiting <https://myaccount.google.com> in a new browser window.
- Review the Recent Security Activity section to confirm whether the alert is genuine.

- Hover over links before clicking to inspect the destination URL.
- Report the email as phishing if it is determined to be fraudulent.
- Enable Multi-Factor Authentication (MFA) for additional account protection.

Email 3 Analysis

Screenshot



Overall Verdict

This email exhibits multiple indicators commonly associated with phishing and online survey scams. The suspicious sender domain, misleading branding, unrealistic promotional offer, and persuasive language suggest that the message is not an authentic communication from Coca-Cola. The email should be treated as malicious until its legitimacy can be verified.

1. Header and Domain Name Discrepancies (Primary Red Flag)

Indicator: The sender appears as Coca-Cola <email_Gep2pQ76g78@opmajvpqjcg.georgs-faescht.com>.

Analysis: Legitimate promotional emails from Coca-Cola are typically sent from official company domains such as @coca-cola.com. The sender's email address in this message uses an unrelated domain and contains long, random-looking strings that are uncommon in legitimate corporate communications.

Risk: An unrelated sender domain may indicate an attempt to impersonate a trusted brand and deceive recipients into interacting with the email.

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2. Brand and Context Misalignment

Indicator: The email claims that Coca-Cola is giving away a portable blender and juicer.

Analysis: The advertised prize does not closely align with Coca-Cola's primary products or typical promotional campaigns. The physical address in the footer also does not correspond with Coca-Cola's known corporate locations.

Risk: Brand impersonation is commonly used to increase credibility and encourage recipients to trust fraudulent promotions.

3. Social Engineering Techniques

Indicator: The email contains phrases such as "You have been chosen," "For FREE!" and "GET STARTED NOW!"

Analysis: These statements are designed to trigger excitement, curiosity, and urgency, creating a false sense of exclusivity while emphasizing that the offer is free and quick to claim.

Risk: Users may act impulsively without checking the sender or verifying the promotion through Coca-Cola's official website.

4. Suspicious Call-to-Action

Indicator: The email encourages recipients to click the "GET STARTED NOW!" button.

Analysis: Unexpected promotional emails requesting immediate interaction should always be approached with caution. The button could redirect users to a fake promotional website designed to collect sensitive information.

Risk: Victims may unknowingly provide personal or financial information to attackers.

Red Flags Identified

- Sender domain is unrelated to Coca-Cola.
- Display name can be easily spoofed.
- Randomized sender address.
- Prize appears unusually generous.
- Promotion does not clearly align with Coca-Cola's typical campaigns.
- Creates urgency through persuasive language.
- Encourages immediate action via a prominent button.
- Footer information appears inconsistent with the claimed organization.

Risk Assessment

Category	Assessment
Threat Type	Phishing / Survey Scam / Financial Fraud
Risk Level	High

Category	Assessment
Potential Impact	Identity theft, financial loss, credential compromise
Confidence Level	High

Recommended Actions

- Do not click the “GET STARTED NOW!” button.
- Verify any promotional offer by visiting Coca-Cola’s official website directly.
- Inspect the sender’s email address carefully rather than relying on the display name.
- Hover over links to examine the destination URL before clicking.
- Report the email as phishing if it is determined to be fraudulent.
- Delete the email after reporting it.

Conclusion

This project provided valuable hands-on experience in recognizing and analyzing phishing emails. By examining three different phishing scenarios, I learned how attackers exploit trusted brands, misleading sender addresses, urgency, curiosity, and attractive offers to manipulate users into revealing sensitive information or interacting with malicious content.

The analysis demonstrated the importance of verifying sender identities, carefully inspecting email domains, avoiding suspicious links, and confirming unexpected requests through trusted communication channels. It also reinforced the role of security awareness and critical thinking in preventing phishing attacks.

Overall, this project strengthened my understanding of phishing techniques, social engineering tactics, and email security best practices. The knowledge gained will help me identify potential threats more effectively and contribute to maintaining a secure digital environment.